

SUNNY KUMAR

Meteorologist | Climate and Energy Modelling Consultant

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PROFESSIONAL SUMMARY

A highly skilled atmospheric/climate scientist/Meteorologist with expertise in weather modeling and data analysis. My experience includes high-resolution modeling of weather events, bias correction of data using machine learning techniques, Numerical Weather Prediction (NWP) Model, WRF Model, and the development of climate projections. I am adept at using a wide range of programming languages and software tools, including Python, R, MATLAB, PyPSA and SQL, and have a keen eye for detail and a proactive approach to problem-solving.

WORK EXPERIENCE

Climate Compatible Futures (CCF)

Research Manager

Feb, 2026 - Ongoing Gurugram, Haryana, India

- Lead and develop Climate Compatible Futures' (CCF) climate and energy research strategy, shaping organizational research vision and priorities
- Organizing/participating in stakeholder consultations and roundtables
- Conduct in-depth climate and energy research, analysis, and technical documentation to support evidence-based policy engagement.
- Conduct research and strategic analysis on energy transition pathways and industry decarbonization, identifying policy, technology, and financing solutions to support low-carbon development.

Centre for Environment and Energy Development (CEED)

Senior Associate - Energy

Apr, 2024 - Jan, 2026 Gurugram, Haryana, India

- Led modeling efforts for Jharkhand's Just Transition scenarios, focusing on aligning energy system pathways with state and national sustainability goals.
- Developed and managed the State Energy Database for Jharkhand through systematic collection and integration of diverse energy data sources.
- Compiled the state energy balance to support in-depth understanding of energy flows and inform Sustainable Transition and Just Transition objectives.
- Estimated Jharkhand's renewable energy resource potential across various sources—including solar (utility-scale, rooftop, agricultural, and floating), wind, pumped hydro energy storage (PHES), and bioenergy—at the most suitable resolution levels (block/sub-district, district, and state), to inform sustainable energy planning and transition strategies.
- Conduct long-term sector-level energy demand projections using scenario-based analysis to inform energy transition planning.
- Develop and apply energy system models using PyPSA to analyze demand projections, assess transition pathways, and evaluate system-level impacts.
- Performed climate modeling of temperature and precipitation data for both future and historical trends in Chhattisgarh, at district/grampanchayat levels, utilizing CORDEX (RCP 4.5 and 8.5 scenarios) and IMD datasets to support detailed climate analysis and adaptation planning.
- Carry out Modeling, visualization for Climate and Energy System.
- Conducted electricity demand projections to guide energy infrastructure planning and policy development.
- Compiled the state energy balance to support in-depth understanding of energy flows and inform Sustainable Transition and Just Transition objectives.

Energy Analyst - Climate and Energy

Dec, 2023 - Mar, 2024 Lucknow & Ranchi, India

- Responsible for Energy System Modeling and provides quantitative analyses of national/sectoral energy transition pathways.

- Carry out Modeling, visualization for Climate and Energy System Modeling.
- Provide technical support to energy programs, focusing on Just Transition scenarios, ensuring a smooth transition to sustainable energy practices.

Genesis Ray Energy

Senior Associate (Climate Modeling)

📅 Feb, 2022 - Nov, 2023

📍 Gurugram, Haryana, India

- Developed and improved high-resolution modelling techniques utilizing NWP, WRF, and AI/ML algorithms to produce accurate weather data for the renewable energy industry, resulting in a 15% increase in satisfaction ratings.
- Implemented Krigging techniques to improve spatial interpolation of high-resolution climate modeling data, increasing precision by 20% and ensuring more accurate forecasting for vulnerable populations.
- Developed and validated machine and deep learning model utilizing ANN, LSTM and XGBoost algorithms to improve the bias-correction of ERA-5 data, resulting in more closer value to ground truth over previous methods.
- Analyzed and modeled cyclonic events using historical ERA5 data in conjunction with current weather patterns to accurately predict the potential path of future storms, Tropical cyclones, depressions.
- Collaborated with research teams to identify and gather LULC, Soils, Water, EBird, and additional bird sensitivity data sets, leading to the development of comprehensive bird sensitivity maps.
- Collaborated with business teams to deliver climate reporting big data analytics for high-profile Reports and other relevant external documents.
- Carry out Modeling, visualizing and interpretation of future climate projections.
- Provide data analytics to internal stakeholders for drawing point based inferences.

Intern (Climate Modeling)

📅 July, 2021 - Jan, 2022

📍 Gurugram, Haryana, India

- Carry out research for high-resolution weather related modelling using NWP, WRF and AI/ML to generate high quality weather data for the renewable energy industry.
- Climate data acquisition, data wrangling.
- Carry out Modeling, visualizing and interpretation of future climate projections.

EDUCATION

M.Sc. in Atmospheric Science

National Institute of Technology, Rourkela

📅 July 2019 - August 2021

📍 Rourkela, Odisha, India

B.Sc. General (Physics, Chemistry and Maths)

KOTA GOVT. COLLEGE (UOK)

📅 July 2016 - June 2019

📍 Kota, Rajasthan, India

RESEARCH/PROJECT EXPERIENCE

Energy System Modelling

Industry Research/Project Experience

📅 Present

📍 Centre for Environment and Energy Development (CEED)

- Carry out Energy System Modelling using python, PyPSA and other various tools Jharkhand and Uttar Pradesh.
- Renewable Energy Assessment at District and Blocks levels.
- State Energy Database Development.
- Analyzed and Model various Climate data source such as ERA5, TerraClimate, etc. to understand climate change patterns and it effect over energy sector.

Downscaling and Bias Corrected Historical ERA5 Dataset (Solar/Wind) over South Asia for renewable industry

Industry Research/Project Experience



Past



Genesis Ray Energy

- Developed and refined a high-resolution modelling technique for downscaling historical data in the renewable energy industry.
- Contributed to advancing the state-of-the-art in high-resolution modeling and data analysis techniques for renewable energy applications.
- Utilized Kriging and Numerical Weather Prediction (NWP) techniques to achieve greater accuracy in fine resolution modeling.
- Employed advanced statistical and machine/deep learning techniques, such as LSTM, XGBoost, ANN, and stacked models, to perform bias correction on high-resolution downscale ERA5 data.
- Modeling, visualizing and interpretation of future climate projections.
- Provide data analytics services to internal stakeholders, enabling data-driven decision-making and the extraction of point-based insights.

Physical and Transition risks

Industry Research/Project Experience



Past



Genesis Ray Energy

- Identify, plan, and execute continuous improvement in existing and future weather data development (CMIP5, CMIP6 and CORDEX).
- Conducted a thorough analysis and modeling of cyclonic events.
- Utilized historical ERA5 data to understand the patterns and characteristics of past storms.
- Analyzed current weather patterns to determine potential paths of future storms, including tropical cyclones and depressions.

Artificial Intelligence and Machine Learning for prediction of Summer Monsoon Rainfall

Academic Research/Project Experience



Past



National Institute of Technology, Rourkela

- Utilized advanced machine learning and artificial intelligence techniques, including ANN, SVR, CNN, and LSTM, to predict summer monsoon rainfall in homogeneous regions of India.
- Conducted research on Indian Summer Monsoon Rainfall (ISMR) prediction using ML/AI.
- Presented research findings on rainfall prediction using AI and machine learning at international conferences, including the International Symposium on Tropical Meteorology (INTROMET'21)

Simulation Class Project

Academic Research/Project Experience



Past



National Institute of Technology, Rourkela

- Utilized FNL data to improve the accuracy and precision of cyclone tracking in WRF simulations.
- Conducted nested domain simulations using the WRF model to track the paths of cyclones, including MAHA and BULBUL.
- Demonstrated proficiency in designing and implementing WRF model simulations for cyclone tracking and forecasting.

Numerical Weather Prediction Simulation Internship

Academic Research/Project Experience



Past



National Institute of Technology, Rourkela

- Demonstrated a deep understanding of numerical weather prediction (NWP) and the WRF-ARW model.
- Conducted WRF-ARW model simulations to track the path of the MAHA Cyclone and provided critical insights into its potential impacts.
- Analyzed and interpreted WRF-ARW model output to improve the accuracy and precision of cyclone tracking and forecasting.

TECHNICAL SKILLS

Python ● ● ● ● ●

MATLAB ● ● ● ● ●

R ● ● ● ● ●

Fortran ● ● ● ● ●

CDO NCO NCL OpenGrADS NWP-WRF
Bash/Shell Scripting Unix/Linux

ArcGIS QGIS GDAL PostgreSQL
Machine Learning Artificial Intelligence Neural Networks
TensorFlow KERAS WRPLOT HYSPLIT
PyPSA Atlite PyPSA-Earth
Microsoft Office Powerpoint/Word/Excel
Numpy/Pandas/Geopandas/Xarray/Matplotlib/NetCDF4
Statistical Analysis

OTHER DETAILS

 **Soft Skills:**
Statistical Analysis, Systemic working, Eye for detail, Eager to learn and proactive, Resolute, Adept, Management, Road Map Building

 **Interests:**
AI/ML/DL, Numerical Weather Prediction, Cylonic Events, Spatio-temporal Krigging, Geo-data Analysis, Energy Analytics, Renewable Energy, Climate Change, Rainfall, Atmospheric Oscillations, Meteorological data, Monsoonal Extreme Events, Climate Related Disclosures, ESG

 **Hobbies:**
Trekking, Walking, Photography, Painting, Competitive Coding, Cycling, Manga/Manhwa, Anime

 **Languages:**
English ● ● ● ● ●
Hindi ● ● ● ● ●